Deep Siamese Progressive Decomposition Network for Performance Prediction

Abstract

Due to variations in service duration, operating environments, and maintenance histories, aero-engines exhibit significant diversity in their performance degradation patterns. However, traditional performance degradation trend prediction methods are susceptible to interference from dissimilar degradation patterns. To address the issue, this paper develops a novel performance degradation trend prediction method based on the Deep Siamese Progressive Decomposition Network (DSPDN). Firstly, diverse degradation patterns are categorized into different pattern clusters according to their similarity. Intra-cluster degradation patterns form positive sample pairs, while inter-cluster degradation patterns form negative sample pairs. Second, the DSPDN is trained using these sample pairs and then employed to represent the similarity between test data and training data quantitatively, as well as to predict trends through similarity matching. The similarity weights are adaptively learned by the DSPDN, which not only suppresses the interference from dissimilar degradation patterns but also enhances the influence of similar degradation patterns. Additionally, the sequence decomposition layer in the DSPDN progressively decomposes performance data into trend and seasonal components, further enhancing the capability of the DSPDN to measure similarity. Finally, the performance of the developed DSPDN is verified using actual aero-engine operation data.

Keywords:
Aero-engines, performance degradation trend prediction, various degradation patterns, deep Siamese progressive decomposition network, similarity matching

1. Introduction

To reduce flight delays and lower operating costs, airlines must adopt scientifically informed strategies for aero-engine utilization and maintenance scheduling[1],[2]. Relying on reactive maintenance, where aero-engines are serviced only after a failure or excessively using the same aero-engine without proper rotation, leads to multiple aero-engines requiring factory repairs simultaneously. This creates a bottleneck because maintenance vendors often lack the capacity to handle large-scale, concurrent repairs. Such inefficiencies severely compromise an airline's ability to maintain its flight schedules and operational readiness. Accurate prediction of aero-engines' performance degradation trends plays a crucial role in supporting proactive, tiered maintenance planning, making it an urgent priority for airlines aiming to improve operational efficiency and cost effectiveness[3],[4].

The performance degradation trend prediction methods for aero-engines are usually categorized into two classes: model-driven approaches and data-driven approaches[5]. Model-driven approaches attempt to simulate the degradation process through physical modeling, and demand extensive domain expertise in aero-engine design and assembly[6]. For instance, Huang et al. proposed a model-driven approach to forecast performance degradation trends of aero-engines[7]. However, such techniques rely on proprietary technical knowledge typically controlled by original equipment manufacturers, restricting their adoption within other airlines. In contrast, data-driven approaches offer a more accessible alternative by extracting patterns from operational data to forecast performance degradation trends, without requiring detailed knowledge of the aero-engine's internal mechanisms[8],[9]. Data-driven approaches can be further classified into statistical approaches and deep learning approaches. Statistical approaches, such as the auto-regressive models, are often limited by strict conditions that require stationary data with regular sampling intervals, which are rarely met in real-world aero-engine datasets. Given the highly variable operating conditions of aero-engines and the frequent occurrence of missing or irregular data, traditional statistical models struggle to deliver reliable predictions, highlighting the growing relevance of deep learning-based solutions for this application[10],[11].

Deep learning approaches are gaining substantial traction in the field of mechanized equipment performance monitoring due to their superior ability to model complex, nonlinear patterns[12],[13],[14]. For example, Wang et al. develop a deep neural network with high-temperature deviation constraints to predict the temperature field of the aero-engine combustion chamber[15]. Despite these advancements, traditional performance degradation trend prediction methods still face major limitations when applied directly to aero-engines. On one hand, the diversity of performance degradation modes across different aero-engines, which is often influenced by the variation of working conditions and environmental conditions, makes it difficult for conventional models to generalize effectively[16]. On the other hand, performance degradation trends often exhibit sequence-level long-term dependencies, promptly leading to the explosion or vanishing of gradients in conventional models during the training phase[17]. The following provides a detailed delineation of the encountered challenges:

Various performance degradation modes: Aero-engine degradation follows various complex modes driven by both internal and external factors. Aero-engines approaching failure typically degrade at a much faster rate than those in earlier stages of wear, and these degradation modes typically exhibit complex nonlinear patterns. Additionally, environmental and

operational differences, such as increased passenger load during holidays, can intensify aero-engine workload and alter degradation behavior compared to normal operating conditions. These variations result in multiple coexisting degradation modes, each with distinct data distributions. As a consequence, performance degradation trend prediction becomes more difficult because overlapping degradation patterns can lead to interference and reduce the accuracy of forecasting models.

Sequence-level long-term dependence: Aero-engine degradation often exhibits sequence-level long-term dependencies due to periodic maintenance cycles. Such maintenance introduces temporal patterns into aero-engine performance, making modeling long-term dependencies critical to accurately capture recurring degradation behavior. However, traditional deep learning models are limited by network structures and are prone to problems, such as vanishing or exploding gradients. These shortcomings hinder the model's ability to retain meaningful temporal features over extended sequences[18].

To address these challenges, we develop a novel performance degradation trend prediction method for aero-engines based on the Deep Siamese Progressive Decomposition Network (DSPDN). This method begins by applying an unsupervised clustering algorithm that maximizes inter-cluster differences and minimizes intra-cluster variance to group degradation data into distinct pattern clusters. Although degradation modes vary, aero-engines with similar usage and maintenance histories tend to exhibit comparable trends, exploited by the clustering step. Positive sample pairs are formed within each cluster, while negative sample pairs are formed across clusters. The deep Siamese architecture is then trained on these sample pairs to serve as a similarity measurement model for performance degradation trends. Once trained, DSPDN compares new test data and stored training data via similarity matching, enabling more accurate predictions by adaptively emphasizing comparable degradation modes and minimizing interference from unrelated ones. For feature extraction, the deep Siamese architecture incorporates a Deep Progressive Decomposition Network (DPDN), which separates time series into trend and seasonal components, simplifying similarity comparison. Additionally, a auto-correlation attention mechanism is embedded to extract sequence-level long-term dependencies more effectively, enabling accurate predictions of aero-engine performance degradation trends.

Moreover, the main contributions of this paper are as follows:

- A novel performance degradation trend prediction framework is introduced, which not only suppresses the interference from dissimilar degradation patterns but also enhances the influence of similar degradation patterns, thereby addressing the challenge of various performance degradation modes.
- A deep Siamese progressive decomposition network (DSPDN) is developed, which can recognize long-term performance degradation trends by utilizing learned similarity weights to match test data with relevant historical samples.
- An integrated temporal feature extraction is incorporated within DSPDN, combining a sequence decomposition layer to isolate trend and seasonal components with a sequence-level auto-correlation attention mechanism. They efficiently capture sequence-level long-term temporal dependencies from complex performance degradation trends.
- Experimental results demonstrate the superiority of the DSPDN using real-world operational data.

The remaining sections are listed below. Section 2 reviews some related work. Section 3 elaborates on the DSPDN and applies it to the performance degradation trend prediction for aero-engines. Section 4 shows experimental results on the actual aero-engine dataset. Finally, Section 5 summarizes the main conclusions.

2. Related Work

Aero-engine performance status plays a critical role in determining optimal repair timing, and accurate trend prediction enables airlines to implement tiered maintenance strategies while minimizing the risk of flight incidents. Predicting degradation trends not only extends aero-engine life but also supports preventive maintenance planning, thereby improving operational efficiency and safety. In recent years, deep learning-based prediction models have gained significant attention for their ability to model complex, nonlinear time series. For example, Giorgi et al. introduced a nonlinear autoregressive network to predict exhaust gas temperature (EGT) in military aero-engines under transient conditions[19]. Che et al. enhanced prediction accuracy by integrating attention mechanisms with support tensor regression[20]. Liu et al. improved prediction precision by extracting EGT-related features from flight data using the Enhanced Scale-Aware efficient Transformer (ESAE-Transformer)[21]. Xiao et al. utilized long and short-term memory networks, integrated with the physical topology of the aircraft engine, to predict future EGT sequences, demonstrating their effectiveness in capturing temporal dependencies[22]. Additionally, Fei et al. employed a hybrid long short-term neural network to model exhaust gas temperature margin, successfully capturing its trend under noise conditions[23]. These studies underscore the growing potential of deep learning in overcoming the limitations of traditional approaches, especially in scenarios characterized by complex, multivariate, and irregular time series data.

However, existing deep learning models struggle to capture long-term dependencies across time segments in aero-engine state data, referred to as sequence-level long-term dependencies, due to limitations in their network structures[24]. The Autoformer model, a recent advancement in deep learning-based additive models, has shown strong performance in complex prediction tasks, such as air pollutant forecasting, evapotranspiration estimation, and short-term traffic flow prediction[25]-[28]. Its success largely stems from its multi-head sequence auto-correlation mechanism, which effectively extracts long-range temporal features and overcomes the information bottlenecks typical in traditional architectures. Despite these strengths, applying Autoformer directly to aero-engine performance degradation trend prediction remains challenging due to multiple overlapping degradation modes. These mixed patterns complicate the model's ability to isolate and learn consistent degradation trajectories, thereby reducing its prediction accuracy for performance degradation trends.

3. Proposed Performance Degradation Trend Prediction Framework

This section begins by outlining the complete framework of the proposed aero-engine performance degradation trend prediction method based on DSPDN, as illustrated in Figure 1. The framework comprises three modules: an unsupervised clustering module that classifies degradation modes, a DSPDN module that quantifies the similarity between different

degradation modes, and a similarity matching module that leverages these measurements to predict trends in performance degradation. In the following sections, a detailed explanation of each module is provided, highlighting its roles and interactions within the overall prediction pipeline.

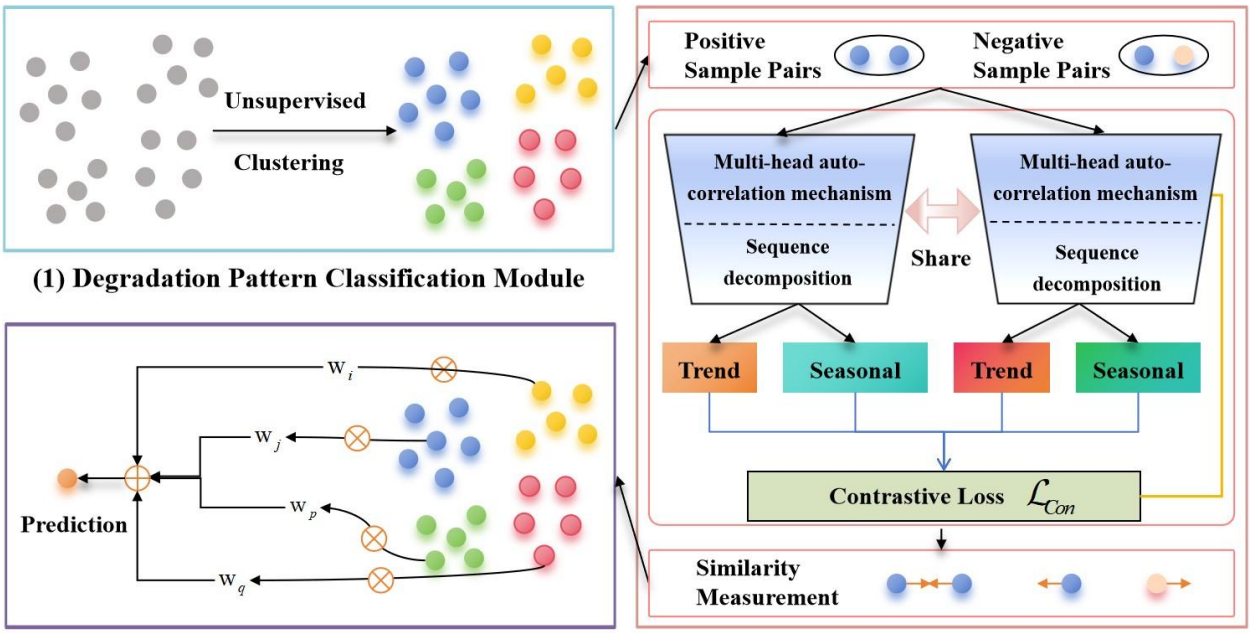


Fig.1 Overall process of the aero-engine performance degradation trend prediction method based on the deep Siamese progressive decomposition network (DSPDN)

3.1. Degradation pattern classification based on unsupervised clustering

The performance degradation modes of aero-engines are exceptionally intricate and varied, rendering it virtually impracticable to undertake rigorous mathematical modeling and pattern classification. In this paper, the primary approach to solving this issue is to employ an appropriate unsupervised clustering algorithm to partition distinct degradation modes into different clusters. Subsequently, trend prediction is carried out through similarity matching to overcome the challenge of predicting with accuracy in the presence of multiple degradation modes.

The aerodynamic performance degradation of aero-engines exhibits high complexity and variability, making it nearly impossible to model or classify patterns using traditional mathematical approaches. To address this challenge, the unsupervised clustering algorithm is used to segment degraded data into different clusters to maximize intra-cluster similarity. This clustering step enables the grouping of similar degradation patterns, thereby reducing interference between differing degradation modes. Once clustering is complete, the similarity matching is used to predict performance degradation trends, which improves prediction accuracy by aligning new data with the most relevant degradation profiles. This method effectively mitigates the difficulty posed by multiple overlapping degradation modes.

Assuming that the training set $\mathbf{X}$ with N samples can be partitioned into N_D distinct clusters, it can be expressed as

$$\mathbf{X} = \bigcup_{i=1}^{N_D} \mathbf{X}^{D_i}$$

where $\mathbf{X}^{D_i}$ represents the i-th cluster. And the objective of clustering is to minimize the summation of the squared distances within each cluster, formulated as

$$\left\{\tilde{\mathbf{X}}^{D_1},\tilde{\mathbf{X}}^{D_2},\cdots,\tilde{\mathbf{X}}^{D_{N_D}}\right\}=\underset{\left\{\mathbf{X}^{D_1},\mathbf{X}^{D_2},\cdots,\mathbf{X}^{D_{N_D}}\right\}}{\arg\min}\left\{\sum_{i=1}^{N_D}\sum_{\mathbf{x}\in\mathbf{X}^{D_i}}\left\|\mathbf{x}-\mathbf{c}_i\right\|_2\right\}$$

where $\mathbf{c}_i$ denotes the cluster center of the i-th cluster. To minimize the aforementioned objective, this paper introduces a latent variable γ_{ij}, which can be represented as

$$\gamma_{ij}=\begin{cases}1, & if\ \mathbf{x}_j\ belongs\ to\ \mathbf{X}^{D_i}\\ 0, & otherwise\end{cases}$$

In this manner, the clustering objective can be converted into the following format:

$$\left\{\tilde{\mathbf{X}}^{D_1},\tilde{\mathbf{X}}^{D_2},\cdots,\tilde{\mathbf{X}}^{D_{N_D}}\right\}=\underset{\left\{\mathbf{X}^{D_1},\mathbf{X}^{D_2},\cdots,\mathbf{X}^{D_{N_D}}\right\}}{\arg\min}\left\{\sum_{i=1}^{N_D}\sum_{j=1}^{N}\gamma_{ij}\cdot\left\|\mathbf{x}_j-\mathbf{c}_i\right\|_2\right\}$$

Obviously, there is a lack of analytical solution to this problem[29]. In this paper, the EM algorithm is employed for its resolution. If the estimated parameters at the t-th iteration are denoted as $\gamma_{ij}^{(t)}$ and $\mathbf{c}_i^{(t)}$, then iterative process for the $(t+1)$-th iteration can be delineated as follows:

E-step: With the current cluster centers fixed, each sample is assigned the cluster to which it belongs, which can be expressed by

$$\gamma_{ij}^{(t+1)}=\begin{cases}1, & if\ i=\underset{p}{\arg\min}\left\{\left\|\mathbf{x}_j-\mathbf{c}_p\right\|_2\right\}\\ 0, & otherwise\end{cases}$$

M-step: With the cluster assignment for each sample fixed, the cluster centers are recalculated and updated accordingly, which can be represented as

$$\mathbf{c}_i^{(t+1)}=\frac{\sum_{j=1}^{N}\gamma_{ij}^{(t+1)}\mathbf{x}_j}{\sum_{j=1}^{N}\gamma_{ij}^{(t+1)}}$$

This process is iteratively repeated until convergence is achieved. Ultimately, a relatively optimal cluster partition can be obtained efficiently[30],[31].

3.2. Similarity measurement of performance degradation modes via the DSPDN

To mitigate the detrimental impacts of multiple degradation modes, conventional prediction methods typically develop individual prediction models for each mode type, resulting in a highly cumbersome model training process. Conversely, this paper presents a mode-robust performance degradation trend prediction method based on the deep Siamese architecture, which encompasses two crucial components: a feature extraction network and a contrastive loss function. Subsequently, detailed descriptions of these components are furnished.

To mitigate the negative effects of multiple modes on trend prediction, traditional time-series prediction methods often require building separate prediction models for each mode, resulting in a time-consuming and inefficient training process. To overcome this limitation, the trend prediction approach is proposed based on a Siamese architecture that integrates two important components: a feature extraction network and a similarity measurement mechanism. This

unified framework enables the model to learn shared representations across different degradation modes and perform predictions through similarity matching, eliminating the need for multiple models. The following sections provide a detailed explanation of each component and its role in enhancing prediction accuracy under complex, multi-mode conditions.

2.2.1 Feature extraction of performance parameters based on the DPDN

The feature extraction network in the Siamese architecture plays a critical role in determining whether the model can accurately assess the similarity or dissimilarity between two sequences by learning and comparing key temporal features from long-state sequence data. Its effectiveness directly influences the performance of similarity measurement and, ultimately, the accuracy of trend prediction. Traditional time series feature extractors used in Siamese architecture mainly include Recurrent Neural Networks and Transformer series. While Recurrent Neural Networks series are theoretically capable of capturing dependencies across multiple time scales, they often encounter gradient explosion or vanishing issues during long sequences due to their recursive structure[32],[33]. Transformer, on the other hand, can model long-range dependencies but typically does so on a point-by-point basis (i.e., individual time steps), which limits their ability to extract degraded patterns across entire time segments.

To address these challenges, the DPDN is used as the feature extraction component in the proposed Siamese architecture, as shown in Fig.2. DPDN overcomes Recurrent Neural Networks' limitations by using a feedforward structure without cyclic characteristics, effectively avoiding gradient-related training issues. More importantly, it incorporates a sequence auto-correlation mechanism capable of flexibly capturing dependencies across time segments, referred to as sequence-level long-term dependencies, rather than isolated time points. This structure breaks through the temporal feature extraction bottleneck often seen in standard Transformers. Additionally, DPDN decomposes the input sequence into trend and seasonal components before measuring similarity, enabling more granular and meaningful comparisons. The trend term reflects the long-term aero-engine performance trajectory under relatively stable conditions, whereas the seasonal term captures short-term, periodic variations influenced by operational events, such as aero-engine washing or high-load holiday flights.

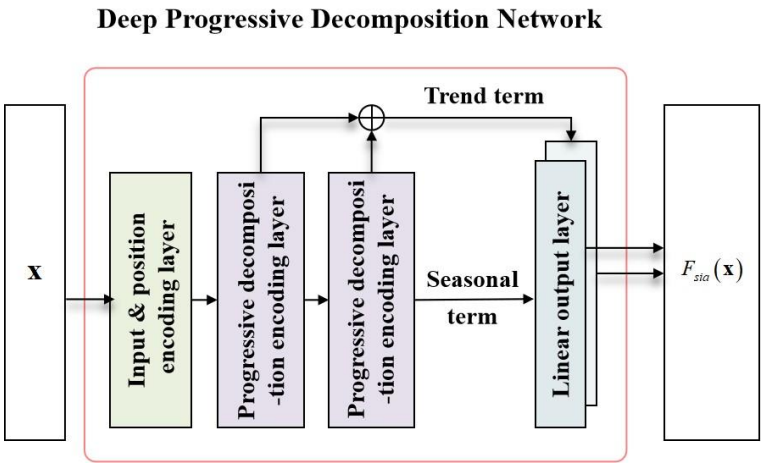


Fig.2 Structure schematic of the DPDN

The DPDN architecture is our model includes an input & position encoding layer, multiple progressive decomposition encoding layers, and two parallel linear output layers. As described in Fig.3, each progressive decomposition encoding layer comprises a multi-head sequence auto-correlation layer, two sequence decomposition layers, and a fully connected feedforward layer. These modular components serve distinct functions. the auto-correlation layer captures

sequence-level temporal patterns, the decomposition layers isolate trend and seasonal dynamics, and the feedforward layer refines the extracted features for final similarity comparison.

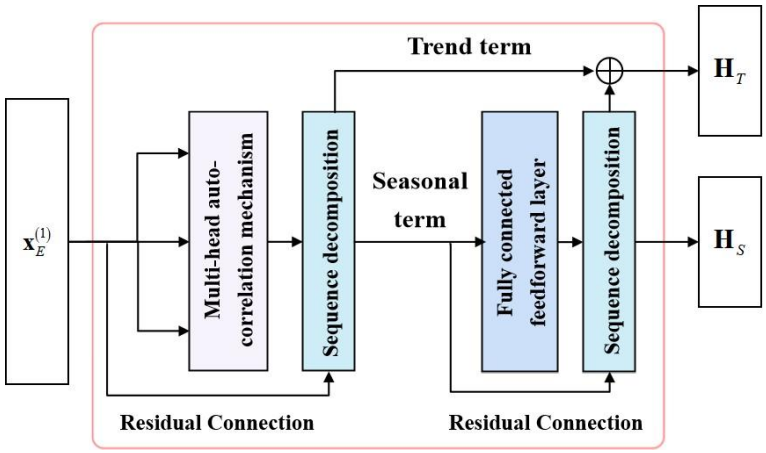


Fig.3 Structure schematic of the progressive decomposition encoder layer

Multi-head sequence auto-correlation mechanism layer: As illustrated in Fig.4 (a), traditional self-attention mechanism captures point-by-point long-term dependencies by computing similarities between individual time steps[34]. In contrast, the sequence auto-correlation mechanism layer within the multi-head sequence auto-correlation mechanism identifies sequence-level long-term dependencies by calculating auto-correlations across entire time segments, as shown in Fig.4 (b). This capability allows the model to capture broader temporal patterns that traditional attention mechanisms often overlook, effectively overcoming the feature extraction limitations of conventional architectures. As a result, the multi-head sequence auto-correlation layer serves as a core component of DSPDN. Structurally, this layer comprises m_A parallel sequence auto-correlation layers, each contributing to a richer representation of time-series dynamics, whose architecture is depicted in Fig.5 (a).

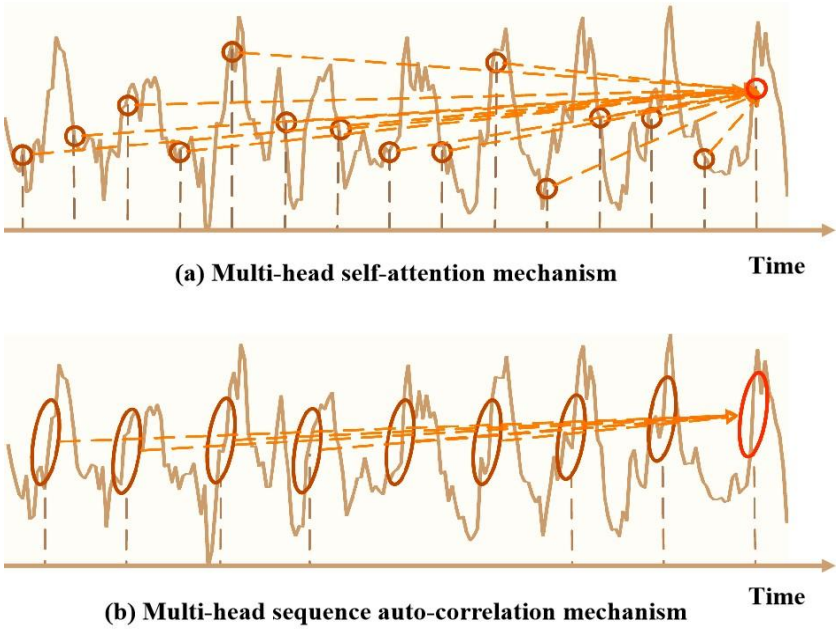


Fig.4 Self-attention mechanism vs series auto-correlation mechanism

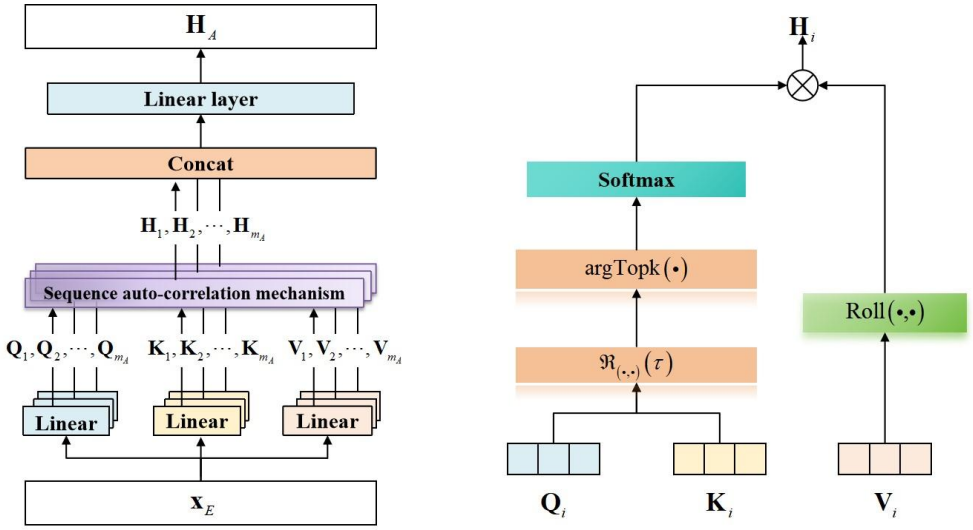


Fig.5 Calculation process of the multi-head series auto-correlation mechanism and series auto-correlation mechanism

Assume the input sequence $\mathbf{x} = \left[x_{ij} \right]_{1 \times d_l}$, where d_l is the sequence length and is, where d_m is the sequence dimension of the encoded data $\mathbf{x}_E = \left[x_{ij}^E \right]_{d_m \times d_l}$ after passing through the input and position encoding layer. We focus on the i-th auto-correlation layers to illustrate the core mechanism.

First, three linear mappings are applied to the input sequence $\mathbf{x}_E$ to generate the query tensor $\mathbf{Q}_i = \mathbf{x}_E \mathbf{W}_i^Q$, key tensor $\mathbf{K}_i = \mathbf{x}_E \mathbf{W}_i^K$, and value tensor $\mathbf{V}_i = \mathbf{x}_E \mathbf{W}_i^V$. Next, as illustrated in Fig.5 (b), the model computes the auto-correlation coefficients $\Re_{(\mathbf{Q}_i, \mathbf{K}_i)}(\tau)$ by evaluating the similarity between the tensor $\mathbf{Q}_i$ and the key tensor $\mathbf{K}_i$, which can be expressed as follows:

$$\Re_{(\mathbf{Q}_i, \mathbf{K}_i)}(\tau) = \frac{1}{d_K} \sum_{t=1}^{d_K} \mathbf{Q}_i(t) \bullet \mathbf{K}_i(t - \tau)$$

where τ represents the delay of the key tensor $\mathbf{K}_i$, and $d_K = \dfrac{d_m}{m_A}$ is the dimension of the key tensor $\mathbf{K}_i$.

Next, the model selects the time delays $\tau_1, \tau_2, \cdots, \tau_k$ corresponding to the top-k of ranking of the sequence auto-correlation coefficient $\Re_{(\mathbf{Q}_i, \mathbf{K}_i)}(\tau)$ (the calculation formula is $k = \lfloor c \times \log(d_l) \rfloor$, where c is a hyperparameter set to 1 according to the literature [25]), which can be expressed as follows:

$$\tau_1, \tau_2, \cdots, \tau_k = \underset{\tau \in \{1,2,\cdots,d_K\}}{\arg\mathrm{Topk}} \left\{ \Re_{(\mathbf{Q}_i, \mathbf{K}_i)}(\tau) \right\}$$

Then, calculate the self-attention weight coefficients $\alpha_1, \alpha_2, \cdots, \alpha_k$, which can be expressed as

$$\alpha_1, \alpha_2, \cdots, \alpha_k = \mathrm{Softmax}\left(\Re_{(\mathbf{Q}_i, \mathbf{K}_i)}(\tau_1), \Re_{(\mathbf{Q}_i, \mathbf{K}_i)}(\tau_2), \cdots, \Re_{(\mathbf{Q}_i, \mathbf{K}_i)}(\tau_k) \right)$$

Finally, the features $\mathbf{H}_i$ captured by the i-th sequence auto-correlation layer can be represented as follows:

$$\begin{aligned}\mathbf{H}_i &= \sum_{i=1}^{k} \alpha_i \cdot \mathrm{Roll}\left(\mathbf{V}_i, \tau_i\right) \\ &= \sum_{i=1}^{k} \alpha_i \cdot [\mathbf{V}_i\left(\tau_i + 1\right), \cdots, \mathbf{V}_i\left(d_V\right), \mathbf{V}_i\left(1\right), \cdots, \mathbf{V}_i\left(\tau_i\right)]\end{aligned}$$

where d_V is the dimension of the value tensor $\mathbf{V}_i$.

Based on the above formula, the self-attention weight coefficients $\alpha_1, \alpha_2, \cdots, \alpha_k$ enable the model to enhance similar time segments while suppressing dissimilar ones. This selective focus allows the network to overcome the limitations of capturing long-range dependencies, effectively identifying sequence-level long-term relationships across extended time intervals. By aggregating features captured by the sequence auto-correlation layers, the model fuses diverse temporal features, allowing the multi-head auto-correlation mechanism to generate a comprehensive and discriminative representation of as the input sequence. This final feature representation $\mathbf{H}_A$ can be expressed as:

$$\mathbf{H}_A = \mathrm{Concat}\left(\mathbf{H}_1, \mathbf{H}_2, \cdots, \mathbf{H}_{m_A}\right)\mathbf{W}^A$$

where $\mathrm{Concat}(\bullet)$ is the concatenation operation of tensors; and $\mathbf{W}^A$ is the parameter matrix of linear mapping.

Sequence decomposition layer: To capture complex temporal patterns in long-term prediction tasks, the sequence decomposition layer decomposes the sequence $\mathbf{H}_A$ into trend term $\mathbf{H}_T$ and seasonal term $\mathbf{H}_S$, and then extracts long-term trends from the trend term $\mathbf{H}_T$ and periodic trends from the seasonal term $\mathbf{H}_S$, which can be represented as follows:

$$\begin{cases}\mathbf{H}_T = \mathrm{AvgPool}\left(\mathrm{Padding}\left(\mathbf{H}_A\right)\right) \\ \mathbf{H}_S = \mathbf{H}_A - \mathbf{H}_T\end{cases}$$

where $\mathrm{Padding}(\bullet)$ is the filling operation of the tensors; and $\mathrm{AvgPool}(\bullet)$ is the average pooling operation of the tensors.

Assuming that DSPDN contains r sequence decomposition layers, the r-th sequence decomposition operation can be expressed as

$$\begin{cases}\mathbf{H}_T^{(r)} = \mathrm{AvgPool}\left(\mathrm{Padding}\left(F_r\left(\mathbf{H}_S^{(r-1)}\right)\right)\right) \\ \mathbf{H}_S^{(r)} = \mathbf{H}_S^{(r-1)} - \mathbf{H}_T^{(r)}\end{cases}$$

where $F_r(\bullet)$ is the nonlinear mapping before the r-th sequence decomposition; $\mathbf{S}_T^{(r)}$ is the trend term after the r-th sequence decomposition; $\mathbf{S}_S^{(r)}$ is the seasonal term after the r-th sequence decomposition.

Then, the trend terms of each sequence decomposition operation are summed to obtain the final sequence decomposition result, which can be expressed as follows:

$$\begin{cases} \mathbf{H}_T = \sum_{i=1}^{r} \mathbf{H}_T^{(i)} \\ \mathbf{H}_S = \mathbf{H}_S^{(r)} \end{cases}$$

where $\mathbf{H}_T$ is the trend term after the progressive sequence decomposition; $\mathbf{H}_S$ is the seasonal terms after progressive sequence decomposition.

Finally, the trend term $\mathbf{H}_T$ and seasonal term $\mathbf{H}_S$ are passed through a linear output layer, and feature concatenation is performed to obtain the output of the feature extraction network, which can be expressed as follows:

$$F_{Sia}(\mathbf{x}) = \left[\mathbf{H}_T \mathbf{W}_T, \mathbf{H}_S \mathbf{W}_S\right]$$

where $\mathbf{W}_T$ is the parameter matrix of the trend output layer, and $\mathbf{W}_S$ is the parameter matrix of the seasonal output layer

The DPDN-based feature extraction network achieves fine-grained similarity measurements by separately comparing the trend and seasonal components of two input samples. This targeted comparison allows the model to distinguish between long-term degradation patterns and periodic variations, thereby enhancing the interpretability and precision of similarity assessments.

Fully connected feedforward layer: To further improve the nonlinear learning ability of DSPDN, the progressive decomposition encoding layer incorporates a fully connected feedforward subnetwork following the sequence decomposition stage. This subnetwork consists of two linear transformation layers separated by a ReLU activation function, enabling the model to capture more complex feature interactions and enhance the expressive power of extracted temporal features. Assume that the output of the fully connected feedforward layer is $\mathbf{H}_{S^*}$, and its calculation formula is as follows:

$$\mathbf{H}_{S^*} = \max\left\{\mathbf{H}_S \mathbf{W}_1^F + \mathbf{b}_1^F, 0\right\} \bullet \mathbf{W}_2^F + \mathbf{b}_2^F$$

Where $\mathbf{W}_1^F, \mathbf{W}_2^F$ are parameter matrices of two linear transformations; and $\mathbf{b}_1^F, \mathbf{b}_2^F$ are bias matrices of two linear transformations.

2.2.2 Similarity modeling of performance parameters based on DSPDN

To accurately measure the similarity of aero-engine performance degradation trends using DSPDN, the model should first be effectively trained. In Siamese architectures, training typically involves inputting a paired sample. The construction of these sample pairs is constructed by clustering the training set $\mathbf{X}$ based on degradation modes, resulting in multiple distinct groups that represent similar degradation behaviors. This clustering step ensures that the model learns to differentiate between intra-cluster and inter-cluster degradation patterns. This process can be expressed as follows:

$$\mathbf{X} = \bigcup_{i=1}^{N_D} \tilde{\mathbf{X}}^{D_i}$$

where $\tilde{\mathbf{X}}^{D_i}$ is the i-th cluster in the relatively optimal partition.

Next, a set of positive sample pairs $\mathbf{X}^{PP}$ is constructed, by selecting two samples from the same degradation mode cluster. Because these samples exhibit similar degradation patterns, pairing them allows the model to learn representations that reflect high similarity within a

cluster. The positive sample set can be expressed as follows:

$$\mathbf{X}^{PP}=\bigcup_{i=1}^{N_D}\left\{\left(\mathbf{x}^{(1)},\mathbf{x}^{(2)}\right):\mathbf{x}^{(1)}\in\tilde{\mathbf{X}}^{D_i},\mathbf{x}^{(2)}\in\tilde{\mathbf{X}}_{r5}^{D_i}\left(\mathbf{x}^{(1)}\right)\right\}$$

where $\tilde{\mathbf{X}}_{r5}^{D_i}\left(\mathbf{x}^{(1)}\right)$ is the companion set of the $\mathbf{x}^{(1)}$ and is randomly selected 5 samples from the set $\tilde{\mathbf{X}}^{D_i}$; $\mathbf{x}^{(1)}$ is any sample in the set $\tilde{\mathbf{X}}^{D_i}$; $\mathbf{x}^{(2)}$ is any sample in the companion set $\tilde{\mathbf{X}}_{r5}^{D_i}\left(\mathbf{x}^{(1)}\right)$.

Then, a set of negative sample pairs $\mathbf{X}^{NP}$ is constructed, where two samples in the sample pairs come from degraded data in different clusters, which can be represented as follows:

$$\mathbf{X}^{NP}=\bigcup_{i=1}^{N_D}\left\{\left(\mathbf{x}^{(1)},\mathbf{x}^{(2)}\right):\mathbf{x}^{(1)}\in\tilde{\mathbf{X}}^{D_i},\mathbf{x}^{(2)}\in\mathbf{X}_{r5}^{-D_i}\left(\mathbf{x}^{(1)}\right)\right\}$$

where $\mathbf{X}_{r5}^{-D_i}\left(\mathbf{x}^{(1)}\right)$ is the companion set of the $\mathbf{x}^{(1)}$ and is randomly selected 5 samples from the set $\mathbf{X}-\tilde{\mathbf{X}}^{D_i}$; $\mathbf{x}^{(1)}$ is any sample in the set $\tilde{\mathbf{X}}^{D_i}$; $\mathbf{x}^{(2)}$ is any sample in the companion set $\mathbf{X}_{r5}^{-D_i}\left(\mathbf{x}^{(1)}\right)$.

Finally, by merging the sets of positive sample pairs $\mathbf{X}^{PP}$ and negative sample pairs $\mathbf{X}^{NP}$, form the final set of sample pairs $\mathbf{X}^{P}$, which can be expressed as follows:

$$\mathbf{X}^{P}=\mathbf{X}^{PP}\bigcup\mathbf{X}^{NP}$$

After constructing the sample pairs, the contrastive loss function is used to train the DSPDN model[35], as illustrated in Fig.6. This loss function guides the model to minimize the distance between the positive sample pairs from the same degradation cluster and maximize the distance between the negative pairs from different clusters.

The contrastive loss $\mathcal{L}_{Con}$ is formally defined as follows:

$$\mathcal{L}_{Con}=\frac{1}{card\left(\mathbf{X}^{P}\right)}\sum_{\left(\mathbf{x}^{(1)},\mathbf{x}^{(2)}\right)\in\mathbf{X}^{P}}\left[\left(1-y_{\left(\mathbf{x}^{(1)},\mathbf{x}^{(2)}\right)}\right)\bullet\left\|F_{Sia}\left(\mathbf{x}^{(1)}\right)-F_{Sia}\left(\mathbf{x}^{(2)}\right)\right\|_2\right.$$
$$\left.+y_{\left(\mathbf{x}^{(1)},\mathbf{x}^{(2)}\right)}\bullet\max\left\{d-\left\|F_{Sia}\left(\mathbf{x}^{(1)}\right)-F_{Sia}\left(\mathbf{x}^{(2)}\right)\right\|_2,0\right\}\right]$$

where $\left(\mathbf{x}^{(1)},\mathbf{x}^{(2)}\right)$ is any sample pair in the set of sample pairs $\mathbf{X}^{P}$; $\left(F_{Sia}\left(\mathbf{x}^{(1)}\right),F_{Sia}\left(\mathbf{x}^{(2)}\right)\right)$ is the embedded representation output of $\left(\mathbf{x}^{(1)},\mathbf{x}^{(2)}\right)$ through the Siamese architecture; If the sample pair $\left(\mathbf{x}^{(1)},\mathbf{x}^{(2)}\right)$ is positive, then the lable $y_{\left(\mathbf{x}^{(1)},\mathbf{x}^{(2)}\right)}$ is set to 0; Otherwise, the lable is set to 1; and the hyperparameter d is set to 2.0 according to the literature [36].

According to the formula, the loss term corresponding to the sample pair $\left(\mathbf{x}^{(1)},\mathbf{x}^{(2)}\right)$ is $\left(1-y_{\left(\mathbf{x}^{(1)},\mathbf{x}^{(2)}\right)}\right)\bullet\left\|F_{Sia}\left(\mathbf{x}^{(1)}\right)-F_{Sia}\left(\mathbf{x}^{(2)}\right)\right\|_2+y_{\left(\mathbf{x}^{(1)},\mathbf{x}^{(2)}\right)}\bullet\max\left\{d-\left\|F_{Sia}\left(\mathbf{x}^{(1)}\right)-F_{Sia}\left(\mathbf{x}^{(2)}\right)\right\|_2,0\right\}$. During the training process, when minimizing the formula, if the $\left(\mathbf{x}^{(1)},\mathbf{x}^{(2)}\right)$ belongs to a positive sample pair, its corresponding loss term can be reduced to $\left\|F_{Sia}\left(\mathbf{x}^{(1)}\right)-F_{Sia}\left(\mathbf{x}^{(2)}\right)\right\|_2$ and tends to

gradually decrease, which means that the distance of the same type of degradation patterns in the metric space becomes smaller and smaller. On the contrary, if the $\left(\mathbf{x}^{(1)},\mathbf{x}^{(2)}\right)$ belongs to a negative sample pair, its corresponding loss can be reduced to $\max\left\{d-\left\|F_{Sia}\left(\mathbf{x}^{(1)}\right)-F_{Sia}\left(\mathbf{x}^{(2)}\right)\right\|_2,0\right\}$, and the loss term also tends to gradually decrease, that is, tends to gradually increase (but not exceed d), which means that the distance of different types of degradation patterns in the metric space is increasing. Importantly, the trained DSPDN does not directly predict trend changes in performance degradation. Instead, it learns a similarity metric that captures how closely a new performance sequence aligns with previously observed degradation patterns.

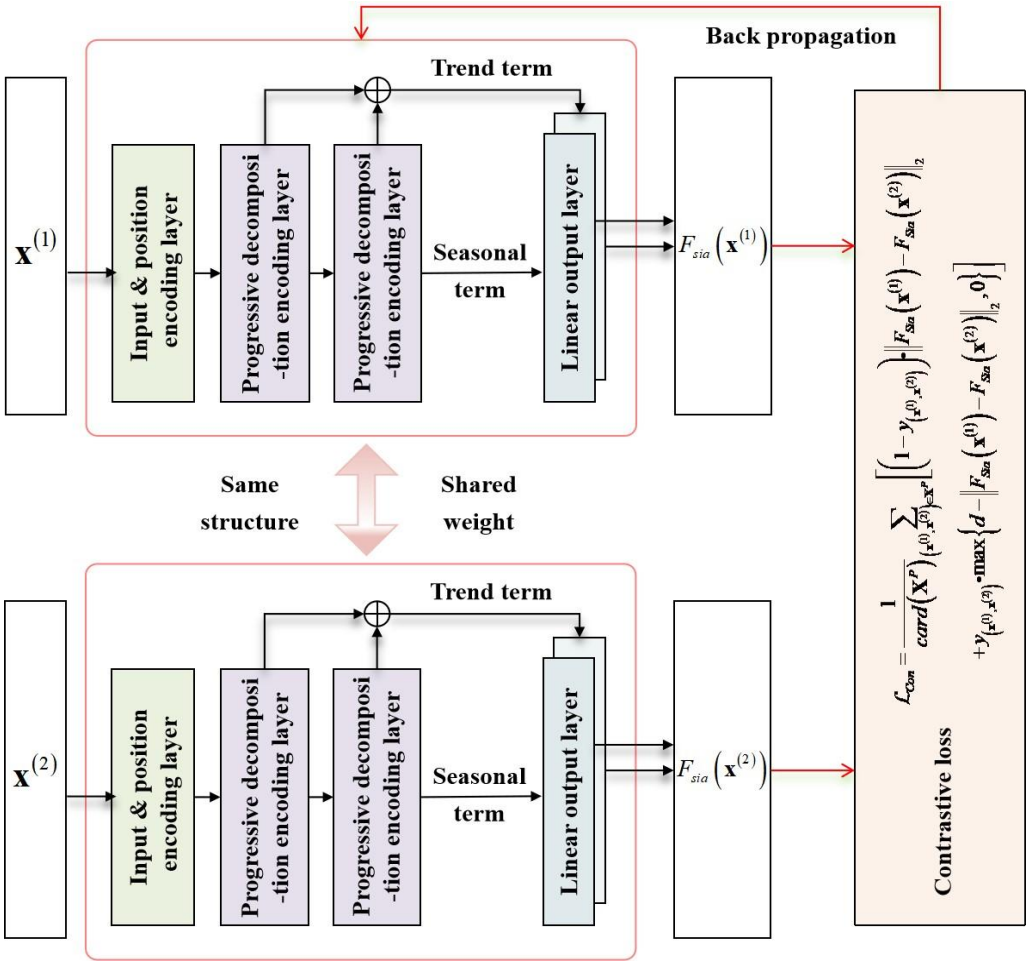


Fig.6 Training schematic of the DSPDN

3.3. Prediction of aero-engine performance degradation trends based on similarity matching

As previously explained, the trained DSPDN maps degraded performance data into an embedded metric space, where the model can evaluate similarity using a simple Euclidean distance. In this space, closer samples indicate stronger similarity in degradation behavior, enabling the model to distinguish subtle differences across various degradation patterns. The similarity $S_M\left(\mathbf{x}^{(1)},\mathbf{x}^{(2)}\right)$ between sample pairs can be defined as follows:

$$S_M\left(\mathbf{x}^{(1)},\mathbf{x}^{(2)}\right)=e^{-\frac{\left[D_E\left(F_{Sia}\left(\mathbf{x}^{(1)}\right),F_{Sia}\left(\mathbf{x}^{(2)}\right)\right)\right]^2}{2}}$$

At this point, for any degraded data to be predicted $\mathbf{u}$, we first calculate the similarity between $\mathbf{u}$ and all samples in the support set $\mathbf{X}$, which can be expressed as:

$$S_M^{\mathbf{u}}=\left\{S_M\left(\mathbf{u},\mathbf{v}\right):\mathbf{v}\in\mathbf{X}\right\}$$

It is worth noting that in the process of similarity matching, samples with higher similarity have stronger representativeness and therefore occupy a greater weight. Assuming the weight of the $\mathbf{v}$ is $w(\mathbf{v})$, it can be expressed as follows:

$$w(\mathbf{v})=\frac{S_M\left(\mathbf{u},\mathbf{v}\right)}{\sum_{\mathbf{t}\in\mathbf{X}}S_M\left(\mathbf{u},\mathbf{t}\right)}$$

Finally, the predicted sequence $\tilde{\mathbf{y}}(\mathbf{u})$ of degraded data $\mathbf{u}$ in the future can be represented as follows:

$$\tilde{\mathbf{y}}(\mathbf{u})=\sum_{\mathbf{v}\in\mathbf{X}}w(\mathbf{v})\bullet y(\mathbf{v})$$

where $y(\mathbf{v})$ the true label for degraded data $\mathbf{v}$.

At this stage, the model can complete trend prediction for any new degraded data sample by comparing it to previously seen patterns in the embedded space. To improve prediction accuracy under diverse degradation conditions, the model assigns similarity-based weights to reference samples. This weighting mechanism reduces the influence of dissimilar degradation modes while emphasizing those with higher similarity, allowing the model to make more reliable predictions across aero-engine air path degradation scenarios.

4. Experimental Results

To verify the effectiveness of the proposed DSPDN-based method in predicting the trend of performance degradation of aero-engines, we conducted experiments using real-world operation and maintenance data collected from a domestic airline. This experimental validation aims to demonstrate the model's practical applicability in real maintenance environments, where accurate trend prediction is crucial for optimizing performance monitoring and maintenance scheduling.

4.1. Experimental data description and analysis

Among various performance indicators, exhaust gas temperature margin (EGTM) serves as a key parameter for assessing the long-term health status of aero-engines. The accurate prediction of EGTM trends enables airlines to maintain flight capability and minimize costs associated with unexpected delays or unscheduled maintenance. In this study, 5 aero-engines with complete degradation histories to ensure data reliability and integrity are selected. To construct the dataset, each aero-engine's installation and disassembly dates from the maintenance reports is first extracted. Using these timestamps, we retrieved the corresponding in-service EGTM data. Therefore, the EGTM sequence $EGTM(p)$ for the p-th aero-engine can be denoted as:

$$EGTM(p)=\left[EGTM_1,EGTM_2,\cdots,EGTM_{M_p}\right]$$

where M_p is the number of flight cycles for the entire service life of the p-th aero-engine.

To ensure the reliability of the experimental results, a leave-one-out cross-validation strategy was adopted. Specifically, for the 5 selected aero-engines, one aero-engine was iteratively designated as the test set while the remaining 4 aero-engines were used as the training set. This process enables individual evaluation of each aero-engine while allowing it to benefit from training on diverse degradation patterns. To generate the input samples, a sliding window approach was applied across each aero-engine's EGTM sequence. The window size was set to L_w (predicted length $L_p = \frac{L_w}{2}$), and the step size of the sliding window L_s was fixed to 10 to ensure sufficient temporal overlap between samples. The q-th sample set for the p-th aero-engine is formally defined as

$$\mathbf{x}_p^{(q)} = \left[EGTM_{10*(q\text{-}1)+1}, EGTM_{10*(q\text{-}1)+2}, \cdots, EGTM_{10*(q\text{-}1)+\frac{L_w}{2}} \right]$$

The label corresponding to the q-th sample is as follows:

$$\mathbf{y}_p^{(q)} = \left[EGTM_{10*(q\text{-}1)+\frac{L_w}{2}+1}, EGTM_{10*(q\text{-}1)+\frac{L_w}{2}+2}, \cdots, EGTM_{10*(q\text{-}1)+L_w} \right]$$

Taking the first aero-engine as the test data, the training set $\mathbf{X}$ and label set $\mathbf{Y}$ corresponding to the training sequence $EGTM_{Tr}(p)$ can be represented as follows:

$$\mathbf{X} = \left[\mathbf{x}_p^{(q)} : 0 < q \le \left\lfloor \frac{M_p - L_w}{L_s} \right\rfloor + 1; p = 2,3,4,5 \right]$$

$$\mathbf{Y} = \left[\mathbf{y}_p^{(q)} : 0 < q \le \left\lfloor \frac{M_p - L_w}{L_s} \right\rfloor + 1; p = 2,3,4,5 \right]$$

The training set $\mathbf{X}_{Te}$ and label set $\mathbf{Y}_{Te}$ corresponding to the test sequence $EGTM_{Te}(p)$ can be represented as

$$\mathbf{X}_{Te} = \left[\mathbf{x}_p^{(q)} : 0 < q \le \left\lfloor \frac{M_p - L_w}{L_s} \right\rfloor + 1; p = 1 \right]$$

$$\mathbf{Y}_{Te} = \left[\mathbf{y}_p^{(q)} : 0 < q \le \left\lfloor \frac{M_p - L_w}{L_s} \right\rfloor + 1; p = 1 \right]$$

In addition, taking $L_w = 360$ (predicted length $L_p = 180$) as an example, the sliding window method was used to construct the number of samples from the historical state data for 5 aero-engines, as summarized in Table 1.

Table 1 Summary of the aero-engine prediction dataset

Situations	Number of the training samples	Number of the test samples	Total number
Case 1	2364	118	2482
Case 2	2364	118	2482
Case 3	2409	107	2516
Case 4	2393	111	2504
Case 5	2462	94	2556

4.2. Evaluation indicators for performance degradation trend prediction results

In time series prediction analysis, researchers commonly evaluate model performance using three key metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). These metrics provide complementary insights, making them particularly useful for interpreting performance across different scales. The mathematical expressions for these evaluation metrics are defined as follows:

$$RMSE = \sqrt{\frac{1}{card\left(\mathbf{X}_{Te}\right)} \times \frac{1}{L_w} \sum_{\mathbf{y}_p^{(q)} \in \mathbf{X}_{Te}} \sum_{j=1}^{L_w} \left(\mathbf{y}_p^{(q)}(j) - \tilde{\mathbf{y}}_p^{(q)}(j)\right)^2}$$

where $\tilde{\mathbf{y}}^{(q)}$ is the predicted value corresponding to the q-th sample $\mathbf{x}^{(q)}$.

$$MAE = \frac{1}{card\left(\mathbf{X}_{Te}\right)} \times \frac{1}{L_w} \sum_{\mathbf{y}_p^{(q)} \in \mathbf{X}_{Te}} \sum_{j=1}^{L_w} \left|\mathbf{y}_p^{(q)}(j) - \tilde{\mathbf{y}}_p^{(q)}(j)\right|$$

$$MAPE = \frac{100\%}{card\left(\mathbf{X}_{Te}\right)} \times \frac{1}{L_w} \sum_{\mathbf{y}_p^{(q)} \in \mathbf{X}_{Te}} \sum_{j=1}^{L_w} \left|\frac{\mathbf{y}_p^{(q)}(j) - \tilde{\mathbf{y}}_p^{(q)}(j)}{\mathbf{y}_p^{(q)}(j)}\right|$$

It can be observed from the above formulas that the smaller the values of evaluation indicators RMSE, MAE, and MAPE, the higher the prediction accuracy of the prediction model.

4.3. Key hyperparameter settings for experimental models

Experiments are implemented in Python 3.9 with PyTorch 1.11.0. The hyperparameters are set according to the recommendations [25] and [37]. The number of clusters in the clustering algorithm $N_D = 20$. The DSPDN feature extraction network includes one input and position encoding layer, two progressive decomposition encoding layers, and two parallel linear output layers. Each progressive decomposition layer contains a multi-head sequence auto-correlation module comprising four parallel auto-correlation layers. After encoding, the input data is set to $d_m = 128$, while the output from the embedding layer has a fixed dimension of 64. The network was trained using the Adam optimizer with a learning rate of 0.0002, a first momentum coefficient $\beta_1 = 0.5$, and a second momentum coefficient $\beta_2 = 0.999$, and a batch size of 128.

4.4. Experimental Comparative Analysis

To verify the prediction performance of our proposed method, four sets of comparative experiments are designed. The first group employed two traditional machine learning methods: Multi-Output Support Tensor Regression (MOSVR) and Multi-Output Decision Tree Regression (MODTR), both implemented using SCIKIT-LEARN with default hyperparameters. The second group focused on recurrent neural network-based approaches, including Recurrent Neural Networks (RNN), long short-term memory networks (LSTM), and Gated Recurrent Unit Networks (GRU). Each was configured with 2 recurrent layers and a linear output layer, with each recurrent layer having a feature dimension of 128. The third comparison involved Transformer-based architecture, including Deep Self Attention Network (DSANet) and Autoformer. DSANet comprises an input and position encoding layer, two Transformer encoding layers, and one linear output layer, using a four-head self-attention mechanism and a

sequence dimension of 128. The Autoformer was implemented using GitHub repository and includes two encoders and one decoder, each with four parallel sequence auto-correlation layers and a sequence dimension of 128. The default hyperparameters were maintained as per the repository. The fourth group used a Similarity-based Prediction (SBP), which computes Euclidean distances between test data and training samples to derive similarity weights for trend prediction. Finally, TreDSPDN is the proposed model in this paper, which leverages DSPDN for performance degradation trend prediction under varying degradation modes.

4.4.1 Experimental results analysis of TreDSPDN model and comparative methods

To comprehensively evaluate the prediction capabilities of different trend prediction methods, each model is evaluated using three different prediction lengths: $L_p = 180$, $L_p = 120$, and $L_p = 60$. When predicting the length $L_p = 180$, Tables 2-4 show the average RMSE, MAE, and MAPE for all methods. Results show that the Autoformer model (from the third comparison group) significantly outperformed traditional machine learning approaches like MOSVR and MODTR, achieving improvements of 17.69% and 42.31% in RMSE, 17.69% and 43.70% in MAE, and 16.52% and 44.48% in MAPE. These gains confirm the superior ability of deep learning architectures to model complex degradation patterns. Autoformer also outperformed recurrent neural network models (RNN, LSTM, GRU) from the second group, with RMSE deductions of 0.24%, 0.47%, and 3.42%, respectively, alongside improvements in MAE and MAPE. This highlights the value of capturing sequence-level long-term dependencies for accurate long-range prediction. Compared with DSANet, Autoformer also showed marked improvement, indicating that its multi-head sequence auto-correlation mechanism extracts temporal dependencies better than traditional Transformer attention. The proposed TreDSPDN method delivered the highest overall performance. In terms of average RMSE, it surpassed all other models, including MOSVR, MODTR, RNN, LSTM, GRU, DSANet, Autoformer, and SBP, by 23.79%, 46.39%, 7.29%, 7.51%, 10.25%, 70.29%, 7.08%, and 15.81%, and achieved the lowest MAE and MAPE across all tested horizons.

Table 2 Average RMSE of different trend prediction methods ($L_p = 180$)

No.	Method	Case 1	Case 2	Case 3	Case 4	Case 5	Average
1	**MOSVR**	5.44	4.82	5.49	4.96	5.16	5.17
	MODTR	6.96	9.20	7.10	6.27	7.23	7.35
2	**RNN**	4.16	4.98	4.37	4.03	3.73	4.25
	LSTM	4.19	5.05	4.31	4.00	3.77	4.26
	GRU	4.39	5.52	4.17	3.88	4.02	4.39
3	**DSANet**	14.29	16.44	11.71	11.7	12.16	13.26
	Autoformer	4.91	4.76	4.38	3.58	3.55	4.24
4	**SBP**	5.05	4.19	5.17	4.40	4.59	4.68
5	**TreDSPDN**	4.26	3.58	4.16	3.58	4.13	**3.94**

Table 3 Average MAE of different trend prediction methods ($L_p = 180$)

No.	Method	Case 1	Case 2	Case 3	Case 4	Case 5	Average
1	**MOSVR**	4.30	3.77	4.36	3.89	4.04	4.07
	MODTR	5.46	7.95	5.70	4.81	5.82	5.95

2	**RNN**	3.40	4.03	3.65	3.12	3.04	3.45
	LSTM	3.42	4.10	3.61	3.10	3.08	3.46
	GRU	3.57	4.58	3.47	3.03	3.30	3.59
3	**DSANet**	13.58	16.00	10.74	11.15	11.52	12.60
	Autoformer	3.90	3.68	3.55	2.86	2.77	3.35
4	**SBP**	4.01	3.31	3.96	3.50	3.57	3.67
5	**TreDSPDN**	3.57	2.88	3.41	2.83	3.34	**3.20**

Table 4 Average MAPE (%) of different trend prediction methods ($L_p = 180$)

No.	Method	Case 1	Case 2	Case 3	Case 4	Case 5	Average
1	**MOSVR**	12.52	11.75	11.74	10.73	11.35	11.62
	MODTR	16.23	25.80	15.51	13.10	16.70	17.47
2	**RNN**	10.21	13.53	9.71	8.17	8.87	10.10
	LSTM	10.30	13.74	9.61	8.14	9.01	10.16
	GRU	10.89	15.32	9.38	8.10	9.73	10.68
3	**DSANet**	40.54	50.99	29.98	31.06	33.23	37.16
	Autoformer	11.44	11.96	9.56	7.80	7.74	9.70
4	**SBP**	11.60	10.52	10.65	9.48	10.00	10.45
5	**TreDSPDN**	10.68	9.36	9.51	7.85	9.72	**9.42**

When predicting at length $L_p = 120$, the average RMSE, MAE, and MAPE across all trend prediction methods are summarized in Tables 5-7. The proposed TreDSPDN method consistently delivered the best performance in all five aero-engine experiments. This superior performance stems from two key design advantages. First, the TreDSPDN uses similarity-based matching, which dynamically adjusts similarity weights to amplify the influence of comparable degradation modes while minimizing the impact of dissimilar ones. This adaptive weighting mechanism mitigates the adverse effects of multiple degradation patterns. Second, the proposed DPDN-based feature extraction network effectively captures sequence-level long-term dependencies, allowing the model to retain the critical temporal structure in the aero-engine degradation process. In terms of average RMSE, the TreDSPDN outperformed MOSVR, MODTR, RNN, LSTM, GRU, DSANet, Autoformer, and SBP by 20.81%, 40.86%, 9.67%, 10.26%, 14.02%, 65.89%, 8.26%, and 9.87%, respectively. Likewise, in terms of average MAE, the proposed method achieved relative improvements of 17.94%, 38.6%, 9.73%, 10.7%, 15.23%, 70.47%, 6.44%, and 5.65%; in terms of average MAPE, the proposed method achieved relative improvements of 16.75%, 39.42%, 12.82%, 13.84%, 18.87%, 70.97%, 4%, and 4.29%, further validating the generalizability of the proposed method.

Table 5 Average RMSE of different trend prediction methods ($L_p = 120$)

No.	Method	Case 1	Case 2	Case 3	Case 4	Case 5	Average
1	**MOSVR**	5.26	5.00	5.34	4.87	5.46	5.19
	MODTR	7.32	7.95	6.15	6.25	7.08	6.95
2	**RNN**	4.58	5.99	4.00	3.77	4.37	4.55
	LSTM	4.60	6.13	3.99	3.79	4.39	4.58
	GRU	4.94	6.57	3.99	3.87	4.53	4.78
3	**DSANet**	9.04	14.61	11.38	12.61	12.59	12.05
	Autoformer	4.90	4.18	4.77	3.93	4.63	4.48
4	**SBP**	4.76	4.08	4.75	4.13	5.11	4.56
5	**TreDSPDN**	4.11	3.57	4.55	3.85	4.49	**4.11**

Table 6 Average MAE of different trend prediction methods ($L_p = 120$)

No.	Method	Case 1	Case 2	Case 3	Case 4	Case 5	Average
1	**MOSVR**	4.12	3.91	4.23	3.82	4.24	4.07
	MODTR	5.48	6.68	4.90	4.72	5.45	5.44
2	**RNN**	3.71	5.07	3.30	2.97	3.46	3.70
	LSTM	3.72	5.22	3.28	2.99	3.47	3.74
	GRU	4.02	5.70	3.23	3.09	3.65	3.94
3	**DSANet**	8.08	14.08	10.58	12.06	11.76	11.31
	Autoformer	3.82	3.36	3.86	3.11	3.69	3.57
4	**SBP**	3.74	3.20	3.61	3.29	3.87	3.54
5	**TreDSPDN**	3.42	2.86	3.73	3.05	3.61	**3.34**

Table 7 Average MAPE (%) of different trend prediction methods ($L_p = 120$)

No.	Method	Case 1	Case 2	Case 3	Case 4	Case 5	Average
1	**MOSVR**	11.89	12.37	11.36	10.45	11.55	11.52
	MODTR	16.15	21.83	13.26	12.91	14.98	15.83
2	**RNN**	11.34	16.86	9.00	8.10	9.70	11.00
	LSTM	11.39	17.34	8.98	8.17	9.76	11.13
	GRU	12.38	18.84	8.97	8.54	10.37	11.82
3	**DSANet**	24.14	44.83	29.32	33.37	33.51	33.03
	Autoformer	10.87	10.31	10.33	8.37	10.09	9.99
4	**SBP**	10.81	10.17	9.61	8.97	10.52	10.02
5	**TreDSPDN**	10.07	9.23	10.25	8.37	10.00	**9.59**

For predicting length $L_p = 60$, the average RMSE, MAE, and MAPE across all trend prediction methods are summarized in Tables 8-10. The TreDSPDN again outperformed all baseline models in terms of average RMSE. In this longer-range prediction scenario, where capturing long-term temporal dependencies becomes even more critical, TreDSPDN demonstrated its scalability and adaptability. It outperformed all comparison methods in average RMSE by 15.83%, 35.77%, 14.59%, 15.48%, 19.52%, 67.68%, 7.97%, and 0.74%, respectively. These results confirm that the combination of DPDN's ability to decompose and interpret long sequences and the Siamese architecture similarity-based prediction mechanism gives TreDSPDN a substantial advantage in forecasting complex real-world aero-engine degradation trends under varying operational conditions.

Table 8 Average RMSE of different trend prediction methods ($L_p = 60$)

No.	Method	Case 1	Case 2	Case 3	Case 4	Case 5	Average
1	**MOSVR**	4.93	4.54	4.87	4.65	5.01	4.80
	MODTR	6.41	7.46	5.95	5.15	6.48	6.29
2	**RNN**	4.65	6.32	3.95	3.79	4.94	4.73
	LSTM	4.69	6.43	3.95	3.80	5.01	4.78
	GRU	5.04	6.89	3.97	3.89	5.31	5.02
3	**DSANet**	12.48	17.23	10.80	9.20	12.79	12.5
	Autoformer	4.67	3.91	4.61	4.08	4.67	4.39
4	**SBP**	4.33	3.82	4.04	3.74	4.42	4.07
5	**TreDSPDN**	3.98	3.81	4.47	3.52	4.43	**4.04**

Table 9 Average MAE of different trend prediction methods ($L_p = 60$)

No.	Method	Case 1	Case 2	Case 3	Case 4	Case 5	Average
1	**MOSVR**	3.81	3.49	3.83	3.57	3.85	3.71
	MODTR	4.85	5.84	4.54	3.95	4.94	4.82
2	**RNN**	3.71	5.36	3.25	3.01	3.96	3.86
	LSTM	3.75	5.48	3.25	3.02	4.05	3.91
	GRU	4.04	5.98	3.22	3.10	4.36	4.14
3	**DSANet**	11.79	16.77	10.17	8.60	12.17	11.9
	Autoformer	3.64	3.15	3.62	3.26	3.57	3.45
4	**SBP**	3.37	2.99	3.08	2.97	3.27	**3.13**
5	**TreDSPDN**	3.20	3.03	3.59	2.88	3.39	3.22

Table 10 Average MAPE (%) of different trend prediction methods ($L_p = 60$)

No.	Method	Case 1	Case 2	Case 3	Case 4	Case 5	Average
1	**MOSVR**	11.03	11.05	10.26	9.66	10.50	10.50
	MODTR	14.18	19.19	12.35	10.72	13.78	14.05
2	**RNN**	11.35	17.92	8.88	8.22	11.42	11.56
	LSTM	11.49	18.29	8.88	8.27	11.69	11.72
	GRU	12.46	19.90	8.93	8.61	12.66	12.51
3	**DSANet**	34.60	53.18	27.85	23.84	34.39	34.77
	Autoformer	10.51	9.87	9.74	8.80	9.86	9.76
4	**SBP**	9.71	9.46	8.20	8.06	8.85	**8.85**
5	**TreDSPDN**	9.29	9.76	9.70	7.85	9.31	9.18

As the prediction length increases, the performance metrics (including average RMSE, MAE, and MAPE) change accordingly, as illustrated in Figs.7-9. Across the three prediction lengths, TreDSPDN consistently ranks first in terms of average RMSE. When the prediction length was either 120 or 180, TreDSPDN consistently secured the top rank in terms of average MAE and MAP, achieving second place only when the predicted length was 60. These results further reinforce the model's trend prediction capability. The growing performance gap between TreDSPDN and other baseline methods as the prediction length increases demonstrates the model's stability and adaptability. This trend supports the conclusion that similarity matching methods perform reliably across varying forecast horizons.

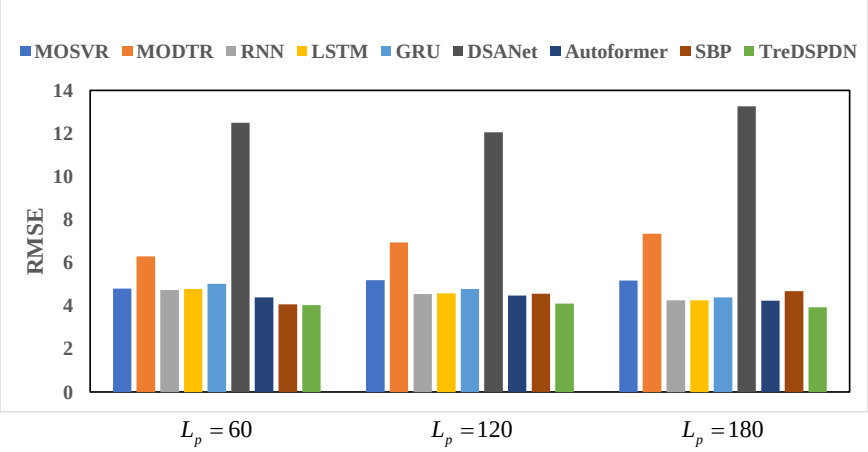


Fig.7 Average RMSE of different trend prediction methods under different prediction lengths

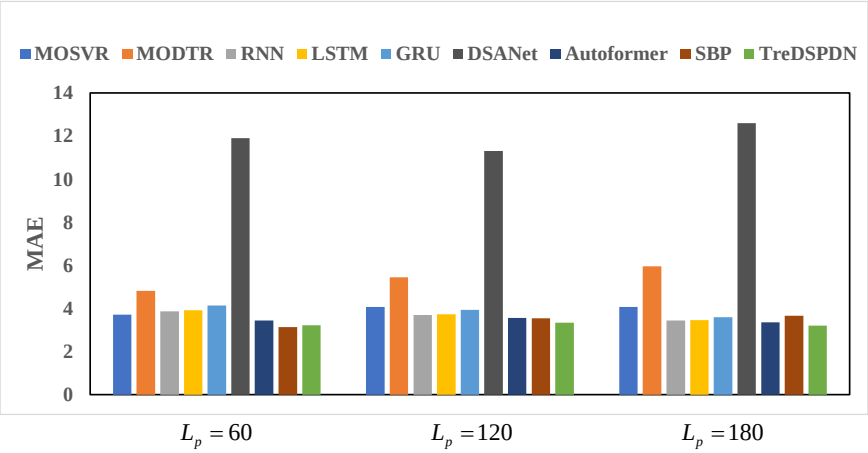


Fig.8 Average MAE of different trend prediction methods under different prediction lengths

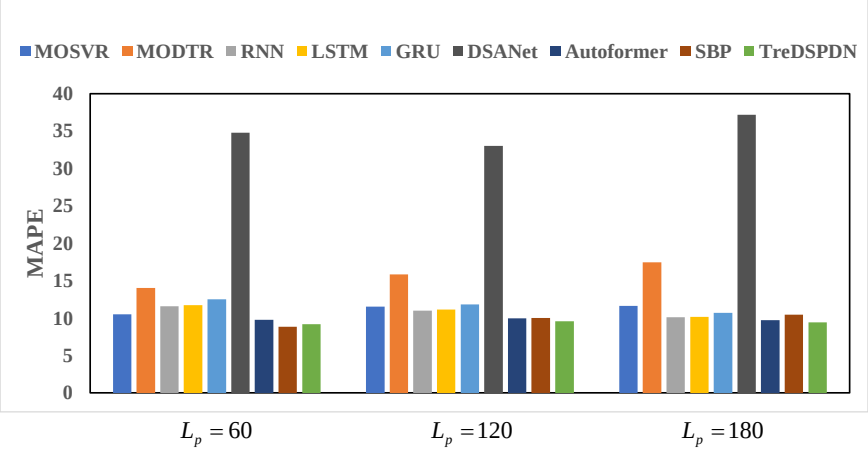


Fig.9 Average MAPE (%) of different trend prediction methods under different prediction lengths

4.4.2 Analysis of the influence of the number of different clustering centers on experimental results

The number of cluster centers is an important hyperparameter that impacts prediction accuracy. Since this parameter governs the granularity of the degradation mode classification, its optimal value significantly affects the downstream similarity-matching process. If too few clusters are used, diverse degradation patterns may be grouped together, reducing intra-cluster homogeneity. Conversely, an excessive number of clusters may scatter similar degradation trajectories across different clusters, weakening the signal in pairwise comparisons. We conducted experiments using various numbers of clustering centers under a fixed prediction length of 180. Table 11 reveals a non-linear trend: TreDSPDN's predictive performance initially improves as the number of cluster centers increases, reaches a peak when the number is set to 20, and then begins to degrade. This behavior underscores the importance of balancing cluster granularity with model generalization. Moreover, across all tested configurations, TreDSPDN consistently outperformed the baseline methods in terms of average RMSE, MAE, and MAPE, thereby confirming the effectiveness of the proposed approach.

Table 11 Average RMSE, MAE, and MAPE (%) of the TreDSPDN using different numbers of clusters

Number of cluster centers	RMSE	MAE	MAPE
5	4.01	3.25	9.55
10	3.96	3.22	9.46
20	**3.94**	**3.20**	**9.42**
40	4.02	3.25	9.54

80	4.01	3.25	9.53
120	4.04	3.26	9.59
160	4.24	3.40	9.97

5. Conclusion

To address the complex and variable degradation trends in aero-engine performance degradation, the novel trend prediction method based on the Deep Siamese Progressive Decomposition Network (DSPDN) is developed. The developed method integrates three key components: unsupervised clustering of degradation patterns, similarity measurement using a deep Siamese architecture, and trend prediction through similarity matching. This framework aims to capture the inherent diversity in aero-engine degradation behaviors and enhance the accuracy of performance degradation trend predictions under multi-modal conditions.

Firstly, recognizing that aero-engines operating under similar service conditions and maintenance regimes tend to exhibit comparable degradation trends, the unsupervised clustering algorithm is employed to classify degradation patterns into different pattern clusters. To measure similarity between these patterns, the measurement of a DSPDN-based architecture is designed that includes a sequence decomposition layer for separating trend and seasonal components of the input data. This decomposition improves the similarity evaluation by isolating meaningful long-term and periodic variations. Furthermore, the model incorporates a multi-head sequence auto-correlation layer, which extracts sequence-level long-term dependencies more effectively than traditional point-wise attention mechanisms, thereby improving temporal feature extraction across extended time horizons.

Second, the trained DSPDN model is applied to assess the similarity between the test data and the training data. Using these similarity scores, the method depicts future degradation trends via similarity-weighted matching. This process not only amplifies the influence of historically similar degradation modes but also diminishes interference from dissimilar patterns, ultimately improving prediction accuracy. The adaptive weighting strategy offers a robust solution to the challenge of overlapping or co-occurring degradation modes, which often mislead traditional predictive models.

Finally, the proposed method is validated using real-world operation and maintenance data from aero-engines. The experimental results show that the proposed method achieved superior performance compared to several baseline models, including MOSVR, MODTR, RNN, LSTM, GRU, DSANet, Autoformer, and SBP. In terms of average RMSE at a selected prediction length, TreDSPDN achieved an RMSE of 3.94 and delivered relative improvements of 23. 79%, 46. 39%, 7. 29%, 7. 51%, 10. 25%, 70. 29%, 7. 08%, and 15. 81%, respectively, over the mentioned models. These findings confirm the effectiveness of DSPDN in capturing degradation dynamics and underscore its practical value in supporting timely and cost-effective aero-engine maintenance decisions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this study.

Data availability

Data will be made available on request.

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